MGMT275 - Product Management in Tech Companies: Product Discovery

Product Design Sprint

App of Choice: CALM

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Background & Context

We began by discussing which app we would like to use. We chose Calm for the following reasons. First, we believe it addresses an important and growing market, mental health and overall well-being. Second, several of us felt a strong personal connection to the problem. It wasn't just intellectually interesting, it was something we genuinely cared about and wanted to contribute to in a meaningful way. Finally, when we asked ourselves, "Which of the three apps would we most want to be the product manager of right now?" Calm was the clear answer for multiple team members.

We started by trying to list out Calm's core Jobs-To-Be-Done (JTBD).

1) Help me fall asleep, 2) Help me de-stress, 3) Make me less anxious, 4) Help me focus, 5) Be a companion.

All of us downloaded the app and tested it ourselves. We explored both the free version and the premium experience through a free trial, to understand the feature set, and overall user experience. We spent some time understanding monetization, and learned that Calm has been relatively successful in both the consumer market and the B2B market. We recognized that we were playing with one of the market leaders in this space. We wanted to think Moonshot and set

our goal not just to compete for marginal gains against existing competitors, but to expand the overall meditation and wellness market itself:)

Sprint Plan

We spent an hour coming up with a structured design sprint process. We tried to balance our desire to spend ample time on all the steps to do a great job with the finite time we could spend given our schedules. We decided to spend a lot of our meeting time on understanding pain points, prioritizing, identifying solutions and mapping our final user journey. We divided up the prototype and deliverables work. We all did user interviews as we thought that was good for learning. (Full Plan in Appendix)

Understanding User Pain Points

We began this process by picking one user journey, which was falling asleep, then divided it into three stages - Opening the app, finding the sleep playlist, and actually falling asleep. For each part, we wrote down the different actions users might take, their needs in that moment and the pain points they are facing. We did this on Mural. We then did a quicker version for the other key user journey, relieving stress. This was a fun divergent thinking exercise.

We then followed on to the convergent thinking process by clustering and summarizing our ideas and looking for the main pain-points. As we were doing this, we realized something crucial, there were two almost independent jobs to be done! One was short term relief - “ Help me fall asleep now”, “ Help me feel less anxious now” and the second was longer term mental health building - “ Help me build a meditation habit”. Both had different needs and pain-points.

The three highest impact pain point categories we found through our user-journey were

1. Difficulty with the long-term habit formation of meditation, sleep routines and other practices good for long-term mental health and wellbeing.
2. High Friction and Low motivation to use Calm short-term relief from stress, anxiety, sleeplessness.
3. Difficulty dealing with dopamine-based distractions when trying to fall asleep or de-stress.

Prioritization

We decided to use the framework of feasibility, customer value, business viability, and usability criteria scores to decide which pain point we wanted to zone in on. (Table in Appendix & Mural)We decided to focus on short term solutions to stress, anxiety and sleeplessness. We thought it was feasible as there were scientific methods, we saw real customer value in helping users with this issue, we saw massive customer acquisition potential with the immediate value giving us potentially a much higher retention rate and finally as long as users open the app

when they are stressed, the rest should be usable. We did secondary research and saw that this was a problem for both customers and there was a huge potential market for it.

Existing Solutions & Ideation

Calm has breathing exercises and SOS support in the app. However they were buried in a midst of a huge number of options and required several steps to find. Too much friction for when a user was stressed. Headspace was also built very similar to Calm.

In terms of the science of short-term stress relief, we found several evidence-based methods including box breathing, mammalian hugs, sensory grounding techniques and there were apps that specialized in this like Wysa, Woebot. However, they had much smaller audiences and the primary reason was that users didn't have the motivation to do the scientifically best option when stressed.

In terms of what most people actually did when they were stressed, it was some form of a quick dopamine hit like scrolling instagram, watching youtube videos, binge watching tv series. None of these actually helped as the dopamine crash that followed typically left people worse off than they were earlier.

We then spent time ideating our own solutions. We once again split up our solution ideas into three stages. (Full solutions lists & some Drawings in Mural & Appendix).

- **Access**: We wanted the access to be low friction and discussed solutions like widgets, watch notifications, voice command, separate app, SOS button. We decided a production version can have multiple entry points.
- **Personalization**: We wanted to understand what was bothering the user, we discussed a menu of options and chatted with a personalized AI avatar. We decided the AI avatar offered better personalization, a wider range of emotions, an effective way to choose between a large catalog of solutions and a little bit of feel good at this stage to motivate people.
- **Solutions**: All our solution ideas fit into two categories, first the quick feel good hit, these included iconic tv show clips, comedy, reassuring messages from a loved one, photos of loved ones, favourite songs, scenic pictures of nature. The second included scientifically proven methods like breathing techniques, sensory grounding, journaling, exercise etc. We decided to do a combination of the two categories to try and address both the friction to open the app in times of stress and to actually leave the app feeling better. So solutions now became two steps, Feel Good Solution & Scientific Solution.

We were now ready to roadmap and storyboard our solution ideas into one complete solution to address the issue of short-term stress relief.

Proposed Solution

1. SOS Button

The objective of the prototype was to design an SOS support feature that enables users to access immediate emotional assistance during moments of distress, with a focus on minimizing friction and response time. To support this, the SOS button was positioned as a highly visible, persistent element on the home screen so users experiencing anxiety, panic, or overwhelm could access support without navigating multiple layers.

We also explored extended accessibility integrations, including a standalone widget and smartwatch triggers. In the latter case, elevated heart rate signals could prompt users to open Calm, shifting support from reactive to proactive.

Our initial concept included an AI-driven mascot designed to create psychological safety through conversational guidance. The mascot would speak in a customizable voice, ask users how they were feeling, and recommend interventions accordingly. Due to prototyping time constraints, this layer was not implemented, and the flow moved directly into emotion identification.

We also evaluated whether exercises should precede emotion selection, recognizing that users in acute distress may struggle to label emotions. This led to the inclusion of an “I don’t know how I feel” pathway, which routes users into generalized grounding interventions. Emotion categories included Frustrated/Angry, Anxious, Scared, Lonely/Neglected, Overwhelmed, Low Self-Esteem, and the non-specific option, enabling both targeted and flexible entry.

2. DOSE Bridge

The second component of the prototype is the DOSE Bridge, designed as a transition layer that moves users from dopamine-driven coping behaviors toward emotional regulation. Research suggests that stress responses often fall into two extremes: low-effort distraction such as doom scrolling, and effort-intensive coping such as mindfulness. During acute distress, users may lack the readiness to engage in structured interventions, creating a behavioral gap.

To bridge this, we conceptualized a stimulation layer that delivers emotional comfort without harmful spiral effects. Proposed elements included personal media from loved ones, short comedy clips, and reassuring mascot prompts. These stimuli align with the DOSE framework: Dopamine (safe stimulation), Oxytocin (connection), Serotonin (regulation), and Endorphins (stress relief).

Due to time constraints, this layer was not built. Instead, emotional states were mapped directly to CBT and breathing interventions, preserving the regulatory intent without the transition experience.

3. Scientific Grounding

The third component focuses on rapid physiological downshift through evidence-based grounding interventions. Once users exit the stimulation or emotion layer, they enter short regulatory sequences designed to reduce emotional and physiological arousal.

Breathing techniques were prioritized as the primary mechanism due to their low cognitive demand and immediate effectiveness. While the original concept included mascot-led guidance, only the exercises themselves were implemented. Additional grounding methods included sensory anchoring, somatic release, and reflective prompts. The regulation journey followed a three-stage structure:

1. **Release:** Discharge emotional intensity
2. **Regulation:** Stabilize through paced breathing
3. **Anchor:** Provide somatic or reflective closure

Each pathway incorporated distinct visual and sensory environments aligned to emotional states, such as storm-to-calm transitions for anger or warm tones for loneliness.

Users were automatically routed into short sequences such as physiological sigh, box breathing, or 4–7–8 breathing. Interventions required minimal interaction and were designed to cognitively "take over," enabling passive regulation.

We also intended to include a post-regulation reflection layer to help users acknowledge triggers or reframe thoughts. This was not implemented due to time constraints but remains a future enhancement.

Prototype Scope

The prototype evaluated whether users would seek in-app support during stress and how the SOS journey influenced emotional outcomes. Specifically, it tested:

- Whether users open Calm when distressed
- Accessibility of SOS entry
- Engagement with the support flow
- Effectiveness of grounding interventions
- Emotional state post-experience

Measures of Success

- **Customer Acquisition:** Check if our short term stress relief interests new customers
- **Finding the feature:** Can users find our solution when stressed.
- **Process Completion:** Do users complete our process to calm down.
- **Habit Formation:** Do users actually consistently use this when acutely stressed

- **Better Retention:** Identify if the users who use this feature will have a better retention.

We are aware that our solution is moonshot and so we focussed a bit more on the assumptions and signals we want to see and haven't defined the actual metrics we want to measure as we are still we need more testing to decide if we should go ahead to build this solution.

User Testing Learnings

User testing revealed that while the SOS concept delivered emotional value, several usability and behavioral gaps emerged across discovery, interaction clarity, and real-world adoption.

- **Help-seeking is not instinctive.**

A key learning was that users do not naturally think of opening a meditation or wellness app in moments of acute stress. Even when moderately overwhelmed, participants indicated they would be more likely to turn to familiar coping mechanisms such as music rather than seek structured in-app support. This highlights a behavioral barrier beyond interface design.

- **SOS discoverability needs strengthening.**

Participants struggled to immediately locate the SOS button and did not strongly associate it with emotional support. The current placement lacked visual dominance within an already content-dense home screen. Additionally, the red color treatment felt overly alarming to some users rather than reassuring.

- **Emotion mapping creates friction.**

Users found it difficult to map their feelings to preset emotional categories. Emotional states often felt overlapping or ambiguous, leading participants to select neutral options such as "I don't know how I feel." This validated the importance of retaining a non-specific entry pathway.

- **Instruction clarity impacts engagement.**

Certain grounding exercises were not immediately intuitive. For example, users did not realize they needed to physically tap elements in sensory activities, and breathing sequences felt confusing without clearer guidance. The absence of audio or sound prompts further reduced immersion and comprehension.

- **Interventions were emotionally effective.**

Despite usability friction, the regulation techniques themselves, particularly breathing exercises, grounding, and acknowledgement prompts, produced measurable stress reduction. Users reported feeling calmer post-experience, validating the scientific grounding of the intervention design.

- **Users expect continuity post-regulation.**

Participants expressed a desire for the experience to extend beyond a single calming activity. They expected follow-up pathways such as music, sleep stories, or longer relaxation content to help sustain the regulated state.

- **Context of use matters.**

The feature was perceived as more useful in transitional decompression moments such as nighttime unwinding rather than during high-pressure work stress, indicating that different entry triggers may be needed across contexts.

Reflection and Next Steps

This sprint generated learnings across four core dimensions: frictionless access, emotional transition, scientific grounding, and post-regulation continuity.

- **Frictionless access emerged as foundational.**

Users struggled to engage with the SOS button, often perceiving it as an emergency or crisis-only feature. This reaction was largely driven by the button's red color treatment and urgency-driven labeling. We recognized that both visual design and copy require recalibration to position SOS as emotionally supportive rather than clinically alarming. A key next step would be to test alternative color systems, language, and placement to evaluate whether discoverability and click intent improve.

- **The DOSE Bridge remains an unvalidated opportunity.**

While the neurochemical framing is grounded in research, we were unable to build and test this transition layer within the sprint timeline. Our intended direction included a mascot-led experience featuring personal media from loved ones, short comedy clips, and reassuring prompts designed to redirect users away from doom-scrolling behaviors. Future iterations would test not only emotional response but also sustained engagement. An additional consideration is content freshness, as repeated exposure could reduce effectiveness due to novelty decay.

- **Scientific grounding validated intervention effectiveness.**

Breathing and CBT-based techniques consistently reduced user stress levels, reinforcing the evidence base behind physiological regulation. However, when instructions or interaction cues were unclear, effectiveness diminished. This highlighted that translating clinical techniques into intuitive product experiences is as critical as the interventions themselves. We would aim to reintroduce the mascot as a guided facilitator, both instructing and demonstrating exercises to reduce ambiguity and cognitive load.

- **Continuity beyond breathing was a major expectation gap.**

Users perceived the journey as incomplete once breathing exercises ended. While immediate relief was achieved, participants wanted guidance on what to do next, whether transitioning into music, sleep stories, or reflective content. This suggests Calm's opportunity lies not only in acute intervention but in designing a seamless pathway from emotional activation to sustained rest and recovery.

What We Would Do Differently

If we were to iterate on this sprint, we would refine our approach across five key areas:

1. **Validate core assumptions earlier.**

We realized that we moved too quickly into solution design without pressure-testing the foundational behavioral assumptions behind the problem. Understanding user intent and help-seeking behavior should precede feature development.
Example: Before building the SOS flow, we would first test whether users actually open Calm during moments of stress and what triggers that action.

2. **Ground design decisions more deeply in proven research.**

While we referenced scientific frameworks, some product decisions could have been accelerated by leaning more heavily on existing UX and behavioral research rather than exploring multiple directions from scratch.

Example: Using established research on emergency and wellness UI patterns to determine optimal SOS placement, color treatment, and copy before prototyping.

3. **Test the testing environment itself.**

We learned that validating the prototype alone is insufficient; the testing conditions also shape outcomes. A more calibrated simulation would have yielded more reliable behavioral insights.

Example: Conducting internal pilot runs to test both the app flow and the stress-induction setup before live user sessions.

4. **Maintain focus on the macro experience.**

During prototyping, we occasionally over-indexed on micro-interactions and feature detailing, which diluted attention from the broader emotional journey we were designing.

5. **Leverage AI more intentionally.**

AI was envisioned as an enabling layer but was not deeply integrated within the prototype due to time constraints. Future iterations would explore AI not just as an interface element but as an orchestration engine.

Example: Using AI to detect emotional tone, personalize grounding pathways, and guide users through interventions dynamically rather than relying on static flows.

Appendix

1. **Sprint Roadmap**

Objective	
1. customers & pain points & user journey	All hands
2. ask the experts	Individual work
3. what to prioritize (problem statement)	All hands
4. dive into existing solutions	Individual work/ then meet All hands
5. new solution	All hands
6. who to test the riskiest assumption	All hands
7. prioritize solution	All hands
8. prototype & roadmap	All hands
9. narrative building	All hands
10. build a prototype	Individual work/ then meet All hands
11. user testing	Individual work/ then meet All hands
12. final deliverable	Split the work
13. sync up for delivering presentation	All hands

2. **Product Sprint Working Board (Mural)**

   https://app.mural.co/t/dataing7514/m/dataing7514/1770342155914/2c074bd6e10850be67ccb7169b252c9fc463ca97?sender=u4b648d9d19a5d63ec9870300

Image of User Journey - Actions, Needs & Pain points from Mural.

3. Pain Point Prioritization

Area to Solve	Feasibility	Customer Value	Business Viability	Usability
Meditation as a long-term habit	7 / 10	8 / 10	9 / 10	7 / 10
short-term solutions to stress	8 / 10	9 / 10	9 / 10	8 / 10
Addressing cheap dopamine distractions	9 / 10	8 / 10	6 / 10	9 / 10

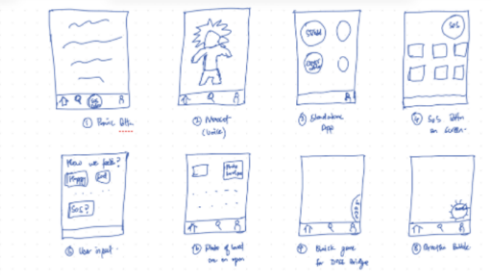
4. Some of our Solution Ideas and drawings

1. **Step 1: Low Design Barrier to Access**
   a. Quick 'SOS' Button on home screen
   b. Easy to access 'Panic Button'
   c. Maybe a lightweight standalone app just for Panic/Stress?
2. **Step 2: The DOSE Bridge**
   a. Uplift to break inertia
   b. Channelize that momentum towards grounding with CBT
   c. Personalized mascot / virtual character to guide
   d. Photos of loved ones
3. **Step 3: Grounding**
   a. Breathing
   b. CBT based physiological methods
   c. Anything else scientific and personalized
4. **Step 4: Reflection?**

!1) Different Colors for Day & Night
2) Character Photo/ Avatar (Celeb/ Personal)
3) Chat & or / Voice
4) Calm Down Button
5) Support Level Scroller
6) Unable to Sleep Similar Setup in dark mode
7) Seperate screen bottom sections and add a immediate solution
8) Standalone app
9) Calm Down section
10) Can't sleep permanent notification
11) Widget
12) Heart Rate tracker - watch
13) Bed time reminder

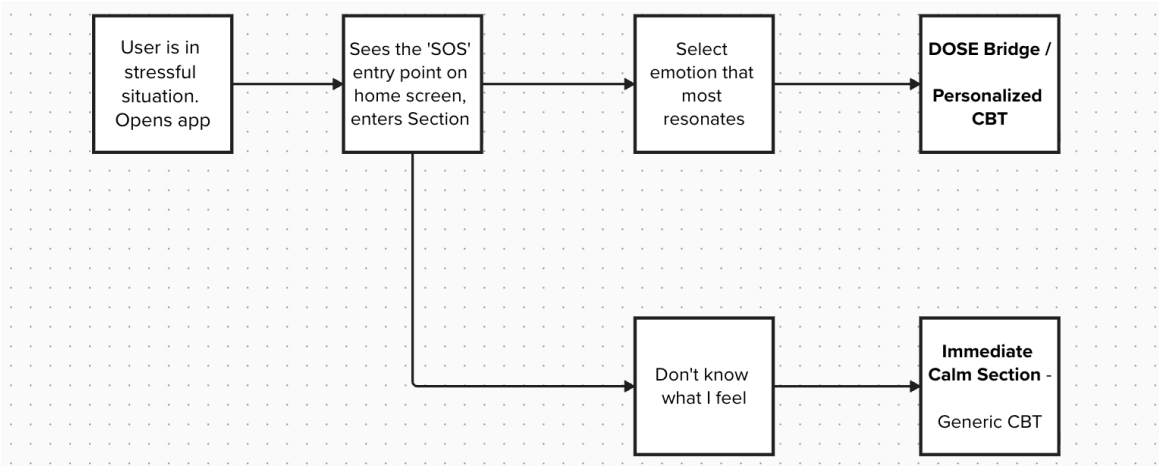
Step 2 Solution Ideas - Pranav:
1. Mascot (Sydney Sweeney/Dua Lipa for me haha) speaking to you to understand your problem
2. Highlight reel of photos of loved ones
3. Quick game for escapism

1. Mascot appears slowly - gentle animation - floating, breathing, gentle glow maybe?
2. Deepens emotional safety - breathes with you, sits beside breathing circle
3. Guides tiny actions - sip of water, unclench, roll shoulders - can even do it with you (visuals)
4. Then closest emotion and maybe specific exercises
5. reflection



Solutions to Calm down
1) Photos of loved one
2) Reassuring message - Pre-added/ text/ voice
3) Comedy show/ TV Show Clip
4) Puzzle/ Exercise
5) Avatar Chat Reassurance
6) Combination
7) Song/ Calming Soundtrack
8) Virtual Hug
9) Scenic Picture of Nature

5. Solution User Journey



6. [Prototype]

7. Scientific Grounding Exercises

Emotion	Step 1: The Release (Immediate)	Step 2: The Regulation (Biological)	Step 3: The Anchor (Closure)	Visual / Sound Theme
Anger & Frustration	**The Burn:** User types rage; taps "Burn." Text shatters into embers with haptic feedback.	**Entrainment Breath:** Dial begins fast (3s inhale / 3s exhale) and slows to (5s / 7s) over 1 min.	**Physical Drop:** Guided shoulder release + prompt "What's the real hurt?"	Stormy Ocean → Still Lake
Anxious & Overwhelmed	**Sensory Anchor:** 5-4-3-2-1 checklist; taps gradually clear blurred forest background.	**Box Breathing:** 4s Inhale · 4s Hold · 4s Exhale · 4s Hold (4 cycles).	**Zen Stone:** User types "One Thing"; text rests on a drifting stone.	Misty Pine Forest (Vertical Rain)
Lonely & Neglected	**The Recall:** Personal photo fades in with Ken Burns zoom + voice note playback.	**Mammalian Hug:** "Hand on heart/belly" prompt; glow pulses with breath.	**Tactile Pulse:** Long-press "Pulse Point" triggers heartbeat haptic.	Warm Campfire (Golden Amber)
"I Don't Know" (Generic)	**Body Map:** User taps tension zones on silhouette torso (glow effect).	**Physiological Sigh:** Inhale · Sip Inhale · Long audible Exhale (3 cycles).	**Gravity Grounding:** 15-second low-frequency "Earth Hum" haptic.	Abstract Silk (Lavender / Indigo)

8. User Interview Notes
9. User Interview Video